

OSSP SKILL Q5:-

1. To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

Code:-

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include <unistd.h>

#include <sys/wait.h>

#define MAX_INPUT 1024

#define MAX_ARGS 100

int main() { char input[MAX_INPUT];

char *args[MAX_ARGS];

printf("===== Mini Shell =====\n");

printf("Type 'exit' to quit.\n\n");

while (1){

printf("MiniShell> ");

if (fgets(input, sizeof(input), stdin) == NULL)

break;

input[strcspn(input, "\n")] = '\0';

if (strlen(input) == 0)

continue;

if (strcmp(input, "exit") == 0)

break;

for (int i = 0; input[i] != '\0'; i++){
```

```

if (input[i] == '\\' && input[i + 1] == ' '){
    input[i] = 1;
    memmove(&input[i + 1], &input[i + 2], strlen(&input[i + 2]) + 1);
}
}

int argc = 0;
char *token = strtok(input, " ");
while (token != NULL && argc < MAX_ARGS - 1){
    for (int i = 0; token[i] != '\0'; i++){
        if (token[i] == 1)
            token[i] = ' ';
    }
    args[argc++] = token;
    token = strtok(NULL, " ");
}
args[argc] = NULL;

pid_t pid = fork();
if (pid < 0) {
    perror("Fork Failed");
} else if (pid == 0) {
    execvp(args[0], args);
    perror("Execution Failed");
    exit(EXIT_FAILURE);
} else {
    wait(NULL);
}

```

```
printf("Shell Closed.\n");  
  
return 0;  
  
}
```

OUTPUT:-

```
#include <stdio.h>  
#include <stdlib.h>  
#include <string.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
#define MAX_INPUT 1024  
#define MAX_ARGS 100  
  
int main()  
{  
    char input[MAX_INPUT];  
    char *args[MAX_ARGS];  
  
    printf("===== Mini Shell =====\n");  
    printf("Type 'exit' to quit.\n\n");  
  
    while (1)  
    {
```

```
apple@apples-MacBook-Pro-2 skill % ./skillq5
```

```
===== Mini Shell =====
```

```
Type 'exit' to quit.
```

```
MiniShell> ls -l
```

```
total 600
```

```
-rwxr-xr-x  1 apple  staff  33360 Aug  5 12:00 skillq1  
-rw-r--r--  1 apple  staff    815 Aug  5 12:01 skillq1.c  
-rwxr-xr-x  1 apple  staff  33416 Aug  5 12:04 skillq2  
-rw-r--r--  1 apple  staff   2265 Aug  5 12:04 skillq2.c  
-rwxr-xr-x  1 apple  staff  34280 Aug 29 15:18 skillq3  
-rw-r--r--  1 apple  staff   7701 Aug 29 15:18 skillq3.c  
-rwxr-xr-x  1 apple  staff  33616 Aug  5 12:11 skillq4  
-rw-r--r--  1 apple  staff   2748 Aug  5 12:09 skillq4.c  
-rwxr-xr-x  1 apple  staff  33672 Aug  5 12:13 skillq5  
-rw-r--r--  1 apple  staff   1715 Aug  5 12:13 skillq5.c  
-rwxr-xr-x  1 apple  staff  33664 Aug  5 12:14 skillq6  
-rw-r--r--  1 apple  staff   1692 Aug  5 12:14 skillq6.c  
-rwxr-xr-x  1 apple  staff  50168 Aug  5 12:16 skillq7  
-rw-r--r--  1 apple  staff   2064 Aug  5 12:16 skillq7.c
```

```
MiniShell> ^
```